

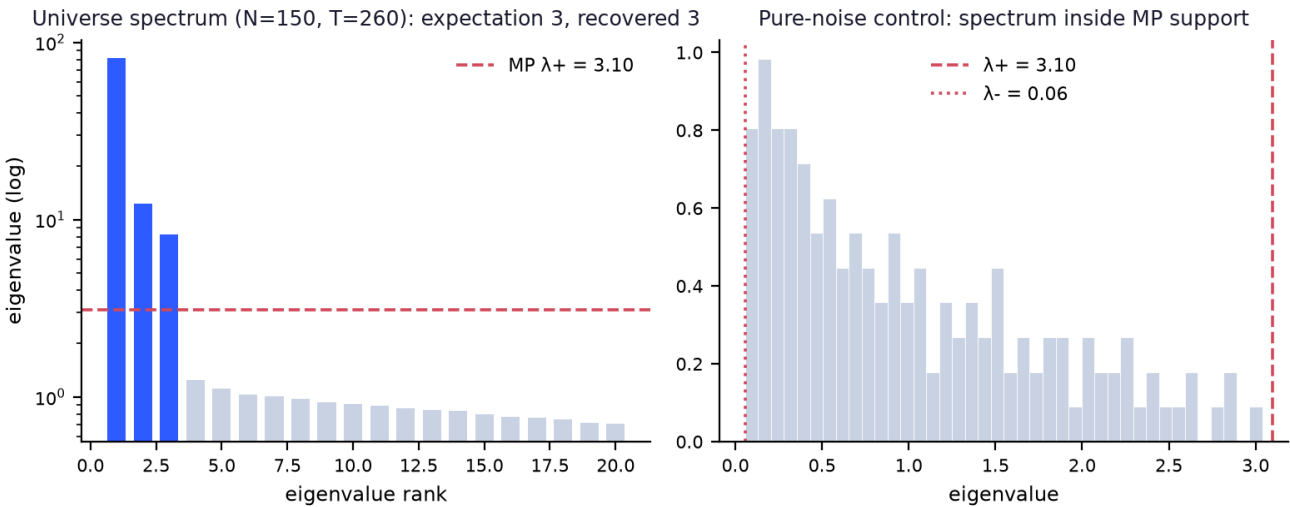
PVE — Stage 5: Risk Model

MP-denoised correlation · factor recovery · risk decomposition · liquidity ladder · repo ~/dev/pve · 2026-07-05

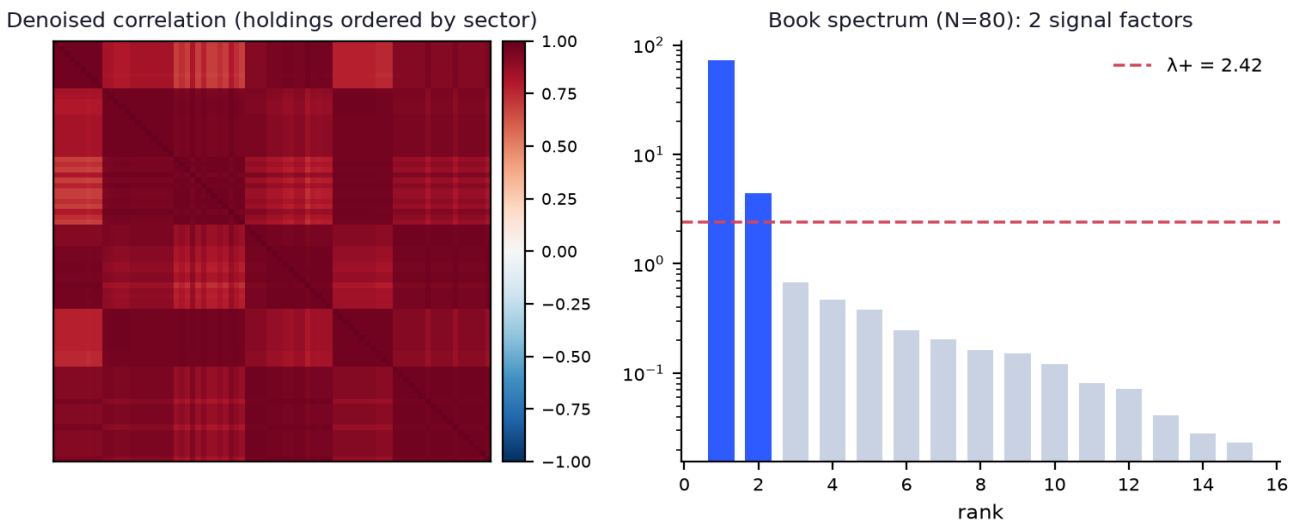
Truth check planted K=3 → recovered 3	MP control noise spectrum inside $[\lambda_-, \lambda_+]$	Condition number $7e+17 \rightarrow 2103$	Effective bets 1.2 of 80 holdings
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The truth check — recovering structure planted five stages ago

Expectation stated first: Stage 0+1 generated the listed universe from a K=3-factor model and logged the truth to the assumptions register. The risk machinery must find ~3 eigenvalues above the Marchenko–Pastur noise ceiling $\lambda_+ = \sigma^2(1+\sqrt{(N/T)})^2$. **Result: exactly 3 recovered** ([81.6, 12.4, 8.3] vs noise bulk ≤ 1.25), the top eigenvector loads 100% positive (the market factor), and the result is stable across seeds. The pure-noise control (right) verifies the bound itself: an iid $N \times T$ matrix's whole spectrum falls inside $[\lambda_-, \lambda_+]$.

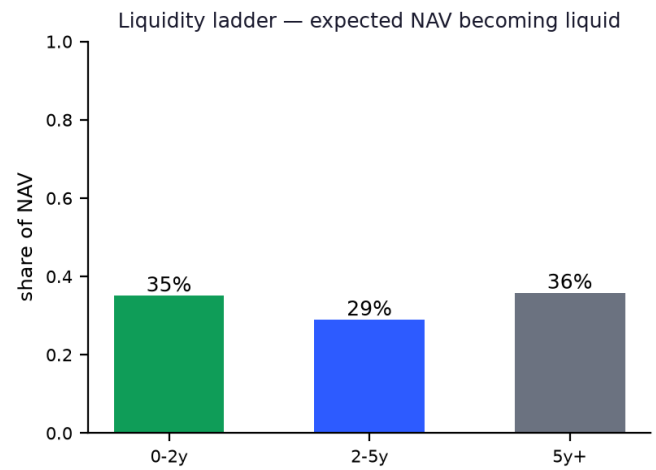
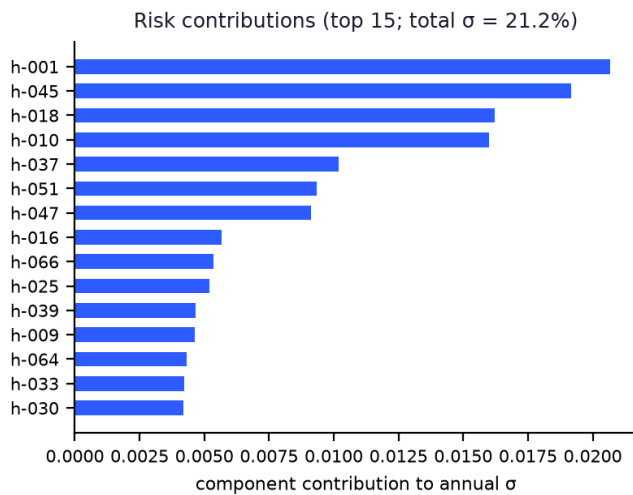


The book-level risk model (80 holdings)



The book itself carries **2 signal factors** — fewer than the universe's 3, and honestly so: each holding's proxy averages ~10 sector comps, which diversifies away idiosyncratic style loadings and leaves the market factor plus one sector contrast. That is a finding about the book (proxy baskets are style-neutralizing), not a failure of the machinery — the machinery just proved itself on the universe above. Denoising: trace preserved at N exactly, unit diagonal, PSD (no projection needed); condition number $7.3e+17 \rightarrow 2103$. Cross-check: Ledoit–Wolf shrinkage ($\delta=0.01$) agrees with the MP-clipped matrix — top-eigenvector $|\cos| = 1.000$, Frobenius distance 1.40.

Risk decomposition, concentration, liquidity



Portfolio σ (annual, denoised) = **21.2%**; component contributions sum to it exactly (Euler). **Effective number of bets: 1.2** (spectrum) / 23.3 (risk-based) against 80 names — the correlation structure means the book holds far fewer independent positions than its holding count suggests. Herfindahl: NAV 0.042, risk 0.043.

Exposure maps (NAV share vs risk share)

dimension	group	NAV	risk
sector	BIOTECH_PHARMA	4.0%	3.5%
sector	CLEANTECH_ENERGY	6.4%	6.4%
sector	CONSUMER	4.3%	4.5%
sector	FINTECH	30.9%	29.4%
sector	HEALTHTECH	2.4%	2.7%
sector	INDUSTRIAL_TECH	6.1%	5.8%
sector	LOGISTICS_MOBILITY	5.9%	5.9%
sector	MEDIA_GAMING	18.8%	20.1%
sector	OTHER	1.9%	1.9%
sector	SOFTWARE_SAAS	19.2%	19.8%
stage	GROWTH	37.5%	36.5%
stage	PRE_IPO	34.3%	34.9%
stage	SEED	0.2%	0.2%
stage	SERIES_A	1.5%	1.5%
stage	SERIES_B	7.7%	7.6%
stage	SERIES_C	18.8%	19.3%
geography	DACH	17.7%	17.3%
geography	NORDICS	34.6%	34.7%
geography	NORTH_AMERICA	30.7%	31.2%
geography	REST_OF_WORLD	0.4%	0.4%
geography	UK_IRELAND	6.2%	6.1%
geography	WESTERN_EUROPE	10.4%	10.3%
vintage	2024	100.0%	100.0%

Judgment calls: proxies are equal-weighted over the Stage-2 peer set (the selector is an unweighted filter); all proxies are market-based so Geltner desmoothing is not applied — the utility exists, is unit-tested, and mark-based series are screened by AR(1). $\sigma^2=1$ for the MP bound (conservative; fitted-bulk mode available in config). Liquidity ladder uses Stage-4 exit-time distributions NAV-weighted. TODO(stage-6): the heatmap, contribution ranking, exposure maps and ladder are the dashboard's risk views.